Daniel Hu

(614) 969-9532 | hu.daniel.w@gmail.com | Github | LinkedIn

Technical Skills

Languages: Python, TypeScript, JavaScript, Java, C++, C, SQL
Frameworks & Data: PyTorch, Scikit-learn, Pandas, OpenCV, Streamlit, React, Angular, Node.js, PostgreSQL
Tools & Platforms: Git, Docker, Linux, AWS (S3, Bedrock, DynamoDB), ROS, Weights & Biases, Vercel, Jupyter
AI Development: Agentic System Design, LLM Orchestration, AI Coding Assistants (Claude Code, Cursor), MCP
Certifications: Career Essentials in Generative AI by Microsoft and LinkedIn; AWS DeepRacer Pro

Experience

Nationwide Columbus, OH
Software Engineering Intern | Nationwide Financial | Project Fulcrum *May 2026 – Present*

- Owned and built a business-critical module end to end (18 features, 22 endpoints) within an enterprise-wide financial platform, designing its PostgreSQL tables, building REST APIs, and an Angular/TypeScript UI
- Met frequently with the business team to gather requirements and verify the implementation matched their needs, translating requirements into structured specs that drove AI-assisted development with GitHub Copilot
- Contributing to a vendor replacement on pace to finish in 1 year and $1M after a prior 5-year, $6M failure
- Built a Black-Scholes pricing engine and Claude-powered competitive intelligence agent (AWS Bedrock, Streamlit) at the internal Enterprise Hackathon, cutting a 2 to 3 day actuarial pricing process to under a minute

Robotics Lab at The Ohio State University (DISL) Columbus, OH
Undergraduate Research Assistant (AI/ML) *Mar. 2024 – Aug. 2024; May 2025 – May 2026*

- Integrated a VLA policy via OpenCV and ROS for autonomous robot manipulation in real-world environments
- Trained CNN-based behavior cloning model to map wrist-camera input to 6-DOF UR5e arm actions
- Reproduced and extended Stanford's UMI environment; reengineered control logic to support Robotiq grippers
- Diagnosed and patched Python and C++ hardware drivers, resolving a critical 100% task failure rate
- Tracked and analyzed training runs with Weights & Biases, improving task completion rate by 62.5% overall

Big Data and Analytics Association Columbus, OH
AI Track Lead *Nov. 2025 – Present*

- Led the Foundations of AI track for a 100+ member org, overseeing teams building AI-powered products
- Mentored students on scoping, AI integration, and presentation; 5 teams presented at the BDAA Research Gala

Projects

Clairvoyant Loss Avoidance & Risk Advisor | *Python, IBM Granite LLM, LangChain* | Repo | Demo 2025

- Won 1st place in the FinTech track at IBM SkillsBuild × OSU Hackathon (100+ Competitors)
- Built an agentic AI portfolio risk platform with real-time hedge proposals and a full audit trail
- Agent autonomously converts global news events into quantitative risk factors and triggers stop-loss alerts
- Powered by IBM Granite LLM; features VaR/ES monitoring and Black-Scholes hedge optimization

Sierra | *TypeScript, React, Node.js, Claude API, OpenAI, Mapbox* | Repo | Live 2025

- AI platform helping engineers route TX transmission lines through ESA, CWA, NEPA, and NHPA compliance
- Engineered a multi-objective A* pathfinder over a 560-node graph, outputting three ranked route options
- Built a 3-phase RAG pipeline over regulatory text; embedded via OpenAI, retrieved via cosine similarity
- Orchestrated a Claude agentic loop with 4 domain tools and MCP search to surface permit timelines

Education

The Ohio State University Columbus, OH
B.S. in Computer Science and Engineering | Statistics Minor | GPA: 3.8 *Aug. 2023 – Dec. 2026*
Engineering Honors Program

- Dean's List (All Semesters), National Merit Scholar, Study Abroad in Berlin (May 2024)
- Coursework: Artificial Intelligence, Software Development, Database Systems, Low-Level Programming, Data Structures & Algorithms, Operating Systems, Data Mining, Machine Learning, Agile Development, Statistics